PhaseViz: A Phase-Aware Post-Match Soccer Analytics Dashboard

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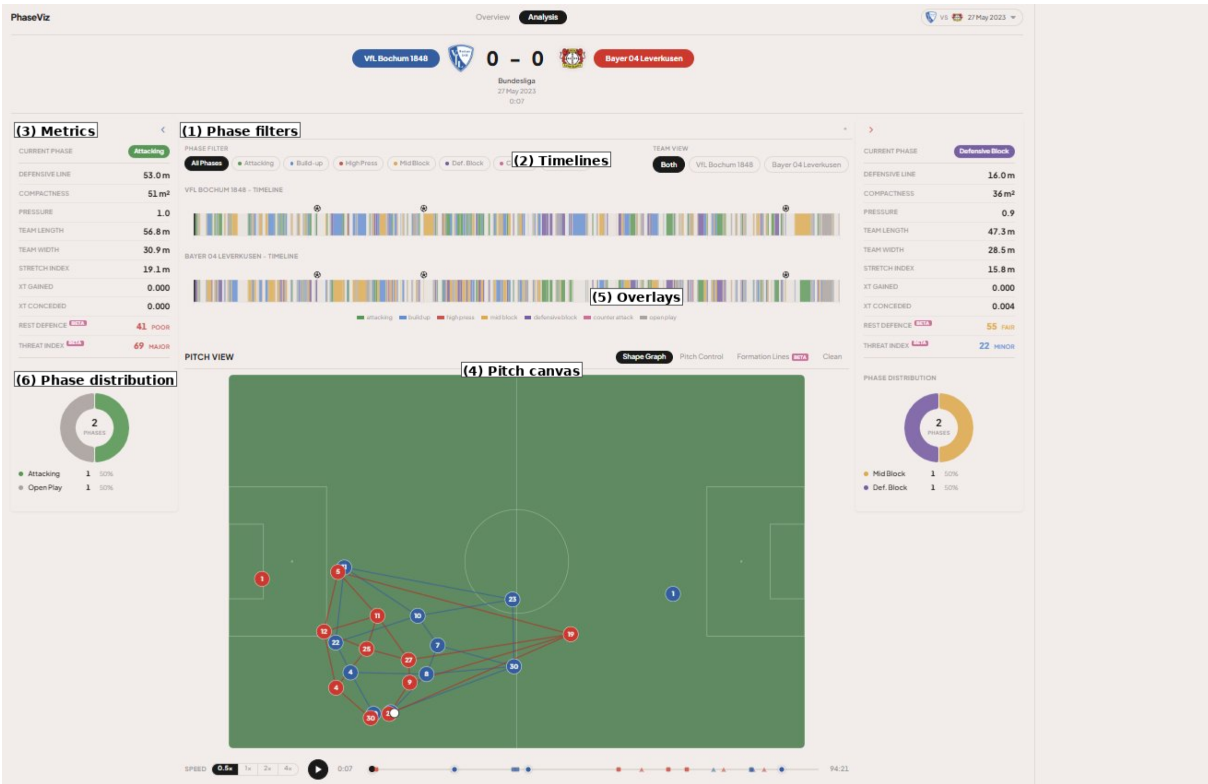

Figure 1: PhaseViz in full analysis mode: (1) phase filter toggles select active tactical phases; (2) dual team timelines show phase-annotated segments with goal markers; (3) metric panels aggregate statistics filtered by the active phase; (4) pitch canvas with shape graph overlay; (5) overlay selector (shape graph, pitch control, formation lines, clean); (6) phase distribution donut chart.

Abstract

This article presents PhaseViz, an interactive dashboard that supports soccer analysts in exploring post-match tracking data through the lens of tactical phases. A soccer match shifts between different situations (pressing, build-up, counter-attacks, defensive blocks), and the metrics that matter change with each. The problem is that current tools present match data as one uniform block, which leaves analysts to mentally segment which statistics apply to which context. PhaseViz addresses this by automatically classifying six tactical phase types from 25 Hz tracking data and filters every view in the dashboard by the active phase: the pitch canvas, the team timelines, the momentum chart, and the metric panels all update together. We describe the data and task abstractions, justify the visual encoding and interaction design, present a case study on a Bundesliga match from the DFL IDSSE dataset [1], and propose an evaluation strategy grounded in Lam et al.'s scenario typology [13].

Keywords: Sports visualization, soccer analytics, tactical phases, coordinated multiple views, formation detection, expected threat.

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1 Introduction

As an invasion game, soccer presents a number of different tactical scenarios, each with different implications. The high number of players on the field makes it a game that demands heavy coordination and benefits from tactical setup arguably more than other invasion sports. A match shifts between different situations, for example: a team can be pressing high up the field to win the ball back, and the other could sitting deep in a compact defensive block, a sudden counter-attack, a patient build-up from the back. Each of these phases has a different spatial signature and a different context that help inform coaching decisions. Post-match analysis, the process through which coaching staff reconstruct what happened and plan adjustments, revolves around understanding these phase transitions.

Modern optical tracking systems now capture every player's position at 25 Hz, producing roughly 135,000 coordinate pairs per match. Simultaneously, event data (passes, shots, tackles) is logged with timestamps and locations. The result is a rich spatiotemporal dataset that, in principle, could reveal how team shape, spatial control, and threat generation vary across tactical contexts.

In practice, however, current tools present this data uniformly. Tools like Wyscout and StatsBomb show tabular summaries and highlight clips (passes, shots, fouls) but lack the ability to place this data in a spatial context and show what led to these events. Tracking viewers animate player movements but lack the tactical annotations that give spatial patterns meaning. As Perin et al. [18] note in their survey of sports data visualization, combining discrete event data and continuous tracking data in a single visualization remains "nearly non-existent." PhaseViz tackles this challenge by presenting a novel way to experience and visualize spatiotemporal data: a phase classifier that runs on tracking data, metric panels that show match statistics like expected goals, expected threat, and pass counts, and the ability to filter all metrics by the selected phase.

We present PhaseViz, a coordinated multi-view dashboard that fills this gap. The system uses tracking data to automatically sort six types of tactical phases. When an analyst chooses a phase, the dashboard reacts in different ways. The animated pitch canvas only shows frames that match, with overlays that can be switched between (shape graph, pitch control, formation lines), the timelines highlight the segments that match, and the xT-based momentum chart, formation comparison glyphs, and phase-filtered metric panels that are re-aggregate to show the same context.

Our contributions are:

1. A coordinated dashboard where selecting a phase filters every view to that context.

2. Visual encodings for spatial metrics (Delaunay-based shape graphs [6], pitch control, pressing heatmaps, xT pass arrows, and xG shot glyphs) all brought together in one dashboard.

3. A case study demonstrating how the system reveals tactical patterns (formation shifts, pressing shapes, threat generation) that are invisible in event-only or tracking-only tools.

2 Background and Related Work

2.1 Soccer Analysis and Visualization

Sports analytics started in turn-based games like baseball, where the data is mostly discrete statistics that map naturally to tables and charts. It then moved into invasion sports like basketball and soccer, where the data is spatiotemporal: player positions captured many times per second across a full match. This kind of data benefits heavily from visualization. Perin et al. [18] organize sports visualization research around three data types (box-score, tracking, meta-data) and note that the intersection of box-score and tracking data remains under-explored. The DFL IDSSE dataset [1], released in 2025 with synchronized 25 Hz tracking and event data for seven Bundesliga matches, partially addresses the survey's call for shared benchmark datasets.

Perin et al. [19] introduced SoccerStories, a system for exploring soccer data through game phases. Their key insight, informed by interviews with four soccer experts (journalists, an Opta analyst, a coach), is that phases provide the optimal semantic level for browsing matches: analysts work under time constraints and need to quickly find the phase that explains a game's outcome. SoccerStories decomposes matches into possession sequences ending in shots and uses piece-wise visual aggregation on a spatial layout for narrative communication. PhaseViz builds on this work: (1) automating phase detection directly from tracking data, going beyond event-only browsing; (2) classifying six distinct tactical types beyond possession sequences; and (3) supporting exploratory analysis through coordinated phase-filtered views.

Other systems address related aspects. Janetzko et al. [9] proposed feature-driven visual analytics of soccer data. Stein et al. [24] combined video and movement data for team sport analysis. MatchPad [15] uses interactive glyphs for real-time sports analysis during live matches. SoccerCPD [11] detects formation change-points but does not link them to spatial overlays. Lin et al. [16] present Omnioculars, a coordinated-view basketball system with linked spatial and statistical panels.

2.2 Tactical Phase Detection

Bauer and Anzer [2] demonstrated that detecting counterpressing requires synchronized positional and event data. Bekkers and Sahasrabudhe [3] proposed rule-based counter-attack detection from event sequences. Praca et al. [20] showed that teams spread out measurably wider when attacking than when defending, which supports the case for separating metrics by phase. Our pipeline combines tracking thresholds with event-based counter-attack rules to classify six phase types per team.

2.3 Formation, Shape, and Threat Models

Bialkowski et al. [5] introduced EM-based formation detection from tracking data. Brandes et al. [6] proposed shape graphs (stable Delaunay subgraphs) as instantaneous representations of team shape, avoiding temporal aggregation artifacts [22]. Bekkers [4] matches player positions to 27 formation templates via linear sum assignment. Singh's xT model [10] trains a Markov chain on a 12×8 pitch grid to estimate scoring probability from each zone; pass threat is the destination value minus the origin.

3 Design

3.1 Data Abstraction

Following Munzner [17], we characterize the data as follows. The primary items are 22 players, one ball, phase segments (1,472 per match), events (1,429 per match), and formations (1,439 per match). Tracking frames (68,377 per match at 25 Hz) provide spatial positions on a continuous field. Links are derived from Delaunay triangulation of outfield players, producing a formation mesh [6]. Using raw positional data we compute per-frame metrics: defensive line height, compactness, team length and width, pressure proxy, expected goals (xG), and expected threat (xT). xG estimates how likely a shot is to result in a goal, a value between zero and one based on shot location, angle, and context. xT measures how much a pass or carry moves the ball into a more dangerous area, using a pre-computed grid of scoring probabilities across the pitch [10]. The key categorical attribute is the phase label, one of six types: attacking, build-up, high press, mid-block, defensive block, or counter-attack. Spatial fields computed from the data include pitch control, a 24×16 grid showing which team would reach each zone first, clipped to the attacking team's convex hull (the smallest shape that encloses all outfield players on that team), and pressing density, a 21×14 heatmap showing where defensive pressure concentrates, computed via kernel density estimation (a smoothing method that turns individual player positions into a continuous intensity surface).

3.2 Task Abstraction

Using Brehmer and Munzner's typology [7], the dashboard is designed to answer five questions analysts typically ask:
Discover: When did the team's tactics change? (Identify phase transitions on the timeline.) **Compare:** How does their shape differ between pressing and defending? (Contrast metrics across phase types.) **Identify:** Where on the pitch are they applying pressure? (Locate pressing zones via heatmap overlay.) **Summarize:** How compact are they during a given phase? (Read aggregated metric cards.) **Locate:** Where are the gaps in their spatial control? (Find pitch control vulnerabilities.)

3.3 Modality and Audience

PhaseViz is a desktop web application for post-match review. The primary audience is match analysts and tactical coaches who review completed matches to identify patterns, prepare scouting reports, and adjust tactical plans. Tory et al. [25] characterize dashboard users as engaging in "data conversations," alternating between overview summaries and drilling into details. Our design supports this: timelines provide temporal overview, the animated

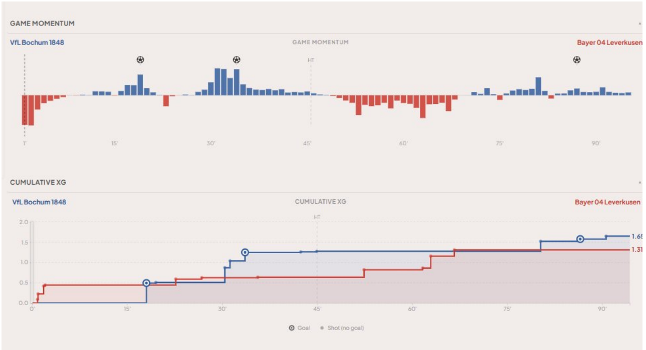

Figure 2: Game momentum and cumulative xG. Diverging bars act as a tug-of-war, showing which team dominates each interval. The line overlay tracks cumulative expected goals. Goal markers distinguish scored goals.

Table 1: Visual encoding summary across dashboard views.

View	Marks	Channels
Pitch canvas	Point, line, area	Position (x,y), hue (team), saturation (xT)
Timeline	Rect	Hue (phase type), position x (time), length (duration)
Momentum	Rect (bar)	Position y (xT), hue (team)
Formation	Point, line	Position (role centroid), edge stability
Metrics	Text, donut	Value (magnitude), hue (category), arc (proportion)
Player xT	Rect	Length (cumulative xT), hue (team)

pitch canvas supplies spatial detail on demand, and phase filtering controls the scope of both. A typical scenario: scan the momentum chart for periods of dominance, click a phase to see its formation and pitch control, then compare team shapes in and out of possession.

3.4 Visual Encoding Design

table 1 summarizes marks and channels. Players appear as colored dots on the pitch, positioned at their (x,y) field coordinates, with color drawn from each team's badge. Team is redundantly encoded in spatial position (home left, away right) [17]. Three selectable overlays add spatial analysis layers:

Shape graph: Delaunay triangulation connects each player to their nearest teammates, with unstable edges (below 45°) removed [6], revealing formation shape and gaps. **Pitch control:** Colored dots on a 24×16 grid showing which team reaches each zone first, clipped to the attacking team's convex hull. **Formation lines:** Lines connecting the defensive, midfield, and attacking lines to show how high or deep each sits and what their shape looks like, notably in a defensive block.

High-xT passes appear as dashed team-colored arrows with pill-shaped labels (fig. 3), while high-xG shots appear as rings scaled by shot quality. These overlays exemplify the integration identified as missing by Perin et al. [18]: event-derived statistics are rendered directly on the continuous spatial canvas alongside the shape graph.

The timeline encodes phases as colored rectangles (hue $\rightarrow$ phase type, length $\rightarrow$ duration) using a muted earth-tone palette. The momentum chart (fig. 2) uses diverging bars (home up, away down) smoothed via bidirectional exponential moving average. Metric panels show phase-filtered aggregates with donut charts for phase distribution.

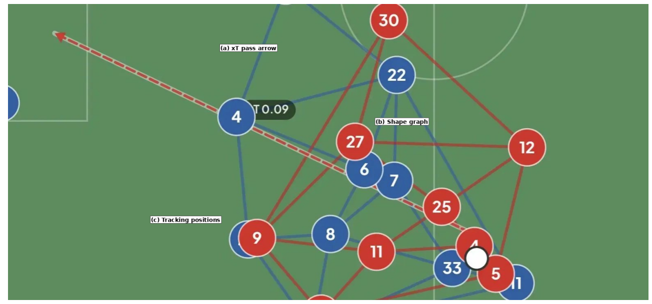

Figure 3: Box-score metrics overlaid on tracking data: (a) a high-xT pass shown as a dashed arrow with threat value label; (b) shape graph edges from Delaunay triangulation reveal formation structure; (c) player positions from 25 Hz tracking. This view combines discrete event statistics with continuous spatial data in a single canvas.

3.5 Interaction Design

Following Dimara and Perin [8]:

Scrubbing: dragging along the timeline animates the pitch, updating player positions, overlays, and metrics frame by frame. **Select:** Clicking a phase segment filters all linked views to that phase, maintaining color consistency [21]. **Navigate:** Playback controls ($0.5\times$ to $4\times$) support temporal exploration. **Filter:** Overlay toggles, team selector, and collapsible panels control the spatial display.

The coordination flow is: phase filter $\rightarrow$ timeline $\rightarrow$ pitch canvas $\rightarrow$ metric aggregation. All views share state through React context. To illustrate: when an analyst selects the *Defensive Block* toggle, only the corresponding segments remain visible on both team timelines, the pitch canvas jumps to only frames from those phases, and the metric panels re-aggregate compactness, defensive line height, and pressure exclusively over those segments rather than the full 90 minutes.

3.6 Animation

The pitch animates at 4 frames per second using Canvas instead of SVG, since SVG struggles with 22 moving player dots each frame. The shape graph is redrawn every frame using `d3-delaunay`. Playback skips frames that fall outside the selected phase filter.

4 Implementation

The system consists of a Python pipeline and a React frontend connected through static JSON.

Pipeline. Nine steps, using kloppy [12] to load the raw Sportec XML. From there, the pipeline computes per-frame features over a 5-second sliding window: defensive line height (average position of the four deepest defenders), compactness (convex hull area), and pressure (how many defenders are within 5 m of the ball carrier). Phases are then classified through two paths: Path A uses tracking thresholds (e.g., high defensive line plus ball in opponent half means high press), and Path B detects counter-attacks from event sequences (possession change followed by rapid forward movement) [3]. Phases shorter than 5 seconds are discarded. The remaining steps compute pitch control (24×16 grid), formations via RoleRep EM [5], shape graphs [6], and EFPI template matching [4] across 27 templates, pressing heatmaps, per-event xT, and finally export everything as JSON.

Frontend. React 18 with D3.js 7. Canvas for animated pitch, SVG for charts. Formation comparison retains edges in $>40\%$ of phases. Period 2 timestamps offset by Period 1 end time.

LLM-assisted development. The system was built with Claude Code (Anthropic) as a coding assistant throughout the project. On the backend, Claude Code implemented the nine-step Python

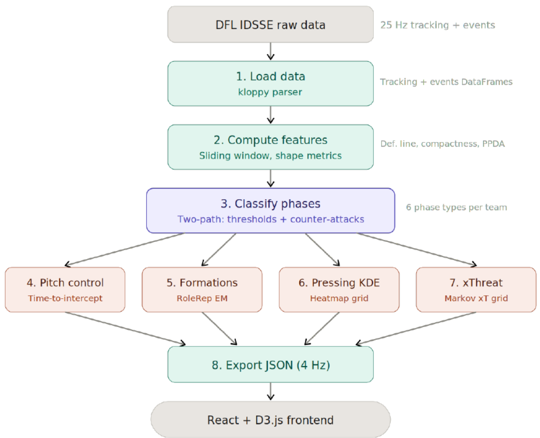

Figure 4: Processing pipeline from raw Sportec XML to frontend JSON.

pipeline, including porting the shape graph algorithm from Brandes et al. and integrating EFPI template matching across 27 formations. It also debugged data issues such as Period 2 timestamp offsets, formation labels producing incorrect results from normalized coordinates, and pitch control file size optimization (320 MB to 3.2 MB). On the frontend, Claude Code built the React/D3.js components and ported the shape graph to JavaScript using `d3-delaunay` for the live overlay. The author directed all analytical goals, chose which algorithms and papers to implement, identified bugs through visual inspection, and drove iterative refinement.

5 Proposed Evaluation

We propose three evaluation strategies based on Lam et al.'s work [13].

Case study (Scenario 7). Match J03WN1 (VfL Bochum vs. Bayer Leverkusen) features a red card at minute 8, forcing formation adaptation from 10 to 9 outfield players, and goals at minutes 18 and 33. Selecting attacking phases on the home team's timeline reveals a 3-4-2-1 in-possession formation with shape graph edges connecting the back three to a diamond midfield. The momentum chart shows a clear xT spike at minute 18 coinciding with the first goal; the phase-filtered metric panel confirms elevated xT gained (0.49) during that phase. Toggling the pitch control overlay during a high-press phase reveals the defending team's convex hull contracting toward their own goal, with the attacking team controlling the majority of grid points inside the hull. These observations, which require integrating phase context with spatial overlays, illustrate the type of insight PhaseViz is designed to support.

Task study (Scenario 2). Give analysts tactical questions to answer using PhaseViz and a standard event-only tool, then compare how quickly and accurately they reach answers. Following Lee-Robbins and Adar [14], the target outcome is *Connect*: relating phases to spatial behavior.

Classification validation. 50 phase segments manually annotated from match video, computing agreement with the automated classifier.

6 Discussion and Future Work

Lessons learned. The design evolved through iterative refinement. An initial hand-crafted xT formula was replaced by Singh's

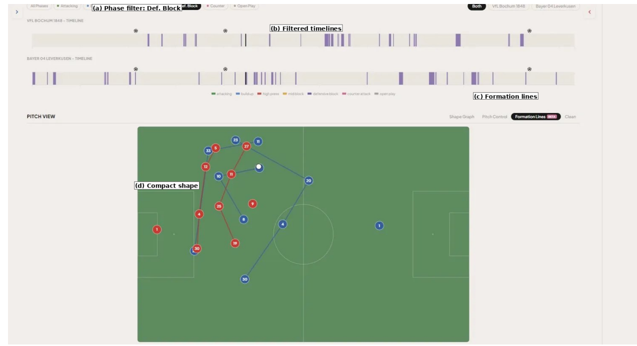

Figure 5: Phase-filtered view demonstrating coordinated filtering: (a) "Def. Block" filter active; (b) timelines show only defensive block segments; (c) formation lines overlay selected; (d) compact defensive shape visible, with players concentrated in their own half.

Markov chain grid after visual inspection revealed unrealistic distributions. An activity-based momentum chart was replaced by pure xT with bidirectional EMA smoothing. Formation labels initially produced incorrect results ("9-1") from centroid-normalized coordinates; switching to raw pitch coordinates resolved this.

Broader implications. Phase-aware filtering is a general pattern: any domain where temporal data can be segmented into meaningful states could benefit from propagating that segmentation across coordinated views. Shape graphs [6] function as a reusable composite mark for team-sport visualization. The system also demonstrates one approach to bridging the box-score/tracking gap identified by Perin et al. [18]: use tracking data to derive segmentation, then aggregate discrete metrics within each segment.

Limitations. No backend; `frames.json` reaches 294 MB. The xT grid was trained on La Liga/Champions League data, not Bundesliga. Phase thresholds are hand-tuned: a 0.02 change in defensive line height shifts detected high-press phases from zero to seventeen. Counter-attack detection found only one instance. Only one of seven matches has been fully demonstrated.

Future work. ML-based phase classification; cross-match comparison; physics-based pitch control [23]; tablet-optimized view for sideline use; including a player-level analysis that enables users to focus on player metrics and to see how they impact team metrics; formal user study with professional analysts.

7 Conclusion

PhaseViz demonstrates that automatic phase classification, combined with coordinated views that all respond to the same phase filter, can structure the exploration of high-frequency soccer tracking data around tactically meaningful units. Building on SoccerStories' insight that phases are the right semantic level for match analysis [19], PhaseViz automates detection from tracking data and adds spatial metrics that only continuous positional data can provide. The result is a dashboard that reveals tactical patterns invisible in event-only or tracking-only tools, and a design pattern applicable to phase-aware temporal analysis beyond soccer.

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References

[1] M. Bassek, M. Raabe, H. Weber, and D. Memmert. DFL IDSSE: An integrated dataset for soccer scouting and evaluation. *Scientific Data*, 12, 2025.

[2] P. Bauer and G. Anzer. Data-driven detection of counterpressing in professional football. *Data Mining and Knowledge Discovery*, 33(5):2009–2049, 2021.

[3] J. Bekkers and A. Sahasrabudhe. Flow of the game: Detecting transitions in soccer. In *MIT Sloan Sports Analytics Conference*, 2023.

[4] J. Bekkers. EFPI: Expected formation position index for soccer. *arXiv preprint*, 2025.

[5] A. Bialkowski, P. Lucey, P. Carr, Y. Yue, S. Sridharan, and I. Matthews. Identifying team style in soccer using formations learned from spatiotemporal tracking data. In *IEEE ICDM Workshop*, pp. 9–14, 2014.

[6] U. Brandes, J. Hiller, and T. Rahlf. Shape graphs for soccer formation analysis. *Journal of Sports Analytics*, 2025.

[7] M. Brehmer and T. Munzner. A multi-level typology of abstract visualization tasks. *IEEE Trans. Visualization and Computer Graphics*, 19(12):2376–2385, 2013.

[8] E. Dimara and C. Perin. What is interaction for data visualization? *IEEE Trans. Visualization and Computer Graphics*, 26(1):119–129, 2020.

[9] H. Janetzko, D. Sacha, M. Stein, T. Schreck, D. Keim, and O. Deussen. Feature-driven visual analytics of soccer data. In *IEEE VAST*, pp. 13–22, 2014.

[10] K. Singh. Introducing expected threat (xT). https://karun.in/blog/expected-threat.html, 2019.

[11] H. Kim, B. Kim, and G. Lebanon. SoccerCPD: Formation and role change-point detection in soccer. In *ACM KDD*, 2022.

[12] kloppy contributors. kloppy: Standardizing soccer tracking and event data. https://github.com/PySport/kloppy, 2024.

[13] H. Lam, E. Bertini, P. Isenberg, C. Plaisant, and S. Carpendale. Empirical studies in information visualization: Seven scenarios. *IEEE Trans. Visualization and Computer Graphics*, 18(9):1520–1536, 2011.

[14] E. Lee-Robbins and E. Adar. Affective learning objectives for communicative visualizations. *IEEE Trans. Visualization and Computer Graphics*, 29(1):1–11, 2023.

[15] P. A. Legg et al. MatchPad: Interactive glyph-based visualization for real-time sports performance analysis. *Computer Graphics Forum*, 31(3):1255–1264, 2012.

[16] Z. Lin et al. Omnioculars: Coordinated visualization for basketball analytics. *Computer Graphics Forum*, 42(3), 2023.

[17] T. Munzner. *Visualization Analysis and Design*. CRC Press, 2014.

[18] C. Perin, R. Vuillemot, C. D. Stolper, J. T. Stasko, J. Wood, and S. Carpendale. State of the art of sports data visualization. *Computer Graphics Forum*, 37(3):663–686, 2018.

[19] C. Perin, R. Vuillemot, and J.-D. Fekete. SoccerStories: A kick-off for visual soccer analysis. *IEEE Trans. Visualization and Computer Graphics*, 19(12):2506–2515, 2013.

[20] G. Praça, F. Clemente, and A. Andrade. Tactical metrics differentiate attacking and defensive phases in soccer. *Journal of Sports Sciences*, 40(12):1378–1390, 2022.

[21] Z. Qu and J. Hullman. Keeping multiple views consistent. *IEEE Trans. Visualization and Computer Graphics*, 24(1):468–477, 2018.

[22] H. Sotudeh. A survey of formation detection in soccer using tracking data. *arXiv preprint*, 2025.

[23] W. Spearman. Beyond expected goals. In *MIT Sloan Sports Analytics Conference*, 2018.

[24] M. Stein, H. Janetzko, et al. Bring it to the pitch: Combining video and movement data. *IEEE Trans. Visualization and Computer Graphics*, 24(1):13–22, 2018.

[25] M. Tory, L. Bartram, B. Fiore, and R. Maciejewski. Dashboard design patterns. *IEEE Trans. Visualization and Computer Graphics*, 2021.